

APPENDIX E

Ephemeris

E.1. Introduction

The ephemerides give the positions and orientations of the planets and minor planets in the solar system. While typically used mostly by the mission-planning groups, they are also needed for attitude control. If a spacecraft must point at a target on a planet or moon, it needs to know the orientation of that body with respect to the inertial frame.

E.2. Planetary orbits

Planetary and moon orbits are necessary for stages of mission when the spacecraft approaches a target. Low-accuracy ephemerides are useful for preliminary work.

High-accuracy tools are available online or can be downloaded from NASA and other sources. A particularly useful high-accuracy tool is the JPL SPICE library [1]. This provides functions to compute the positions and velocities of the planets (including the nonplanet Pluto), the Earth's Moon, and the Sun.

An alternative is to use state vectors from JPL Horizons. JPL Horizon is an online solar-system data and ephemeris computation providing access to key solar-system data. It permits the production of highly accurate ephemerides for solar-system objects including (at the time of this book's publication) 1 124 802 asteroids, 3751 comets, 209 planetary satellites, 8 planets, the Sun, the first and second Lagrange points (L1 and L2), select spacecraft, and system barycenters [2]. Barycenters are one of the focii of an elliptical orbit. Example E.1 shows a nuclear-fusion-propelled spacecraft playing catch-up with I1/'Oumuamua, a recent interstellar visitor. The Horizons state vector that initializes the I1/'Oumuamua state is

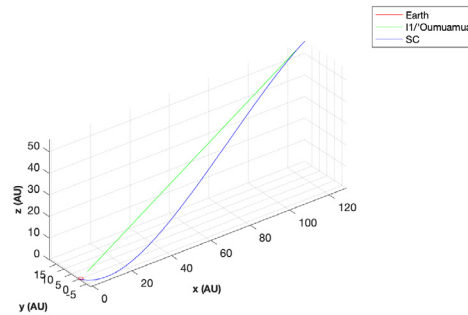
```
jD = 2462576.5; % A.D. 2030-Mar-16 TDB
rA = [ 1.008272975334536E+10; 1.579428200546252E+09;
      4.311503526924462E+09];
vA = [ 2.428440175149515E+01; 3.623455975379303E+00;
      1.063812980138126E+01];
```

The units are km and km/s. The fusion-rocket interception trajectory is found by trajectory optimization. Trajectory propagation uses a simple point-mass Sun gravitational model. The trajectory might be used if one were to design a spacecraft to chase an interstellar object.

```

1
2 %% I1/'Oumuamua from
3 jD      = 2462576.5;% A.D. 2030-Mar-16
4         00:00:00.0000 TDB
5 rA      = [ 1.008272975334536E+10;
6             1.579428200546252E+09; 4.311503526924462E
7             +09];
8 vA      = [ 2.428440175149515E+01;
9             3.623455975379303E+00; 1.063812980138126E
10            +01];
11 mu      = 1.3271e+11;
12 e10     = RV2EI(rA,vA,mu);
13 [r,v]   = RVOrgen(e10,linspace
14             (0,-12*365*86400,200),[],mu);
15 au      = 14959787;
16 Plot3D(r/au)
17
18 % Assume we could launch closer to when asteroid
19 % is detected (2018)
20 e1      = RV2EI(r(:,end),v(:,end),mu);
21 jD0     = jD - 12*365;

```



Example E.1: I1/'Oumuamua and a nuclear-fusion-powered spacecraft playing catch-up with the interstellar object.

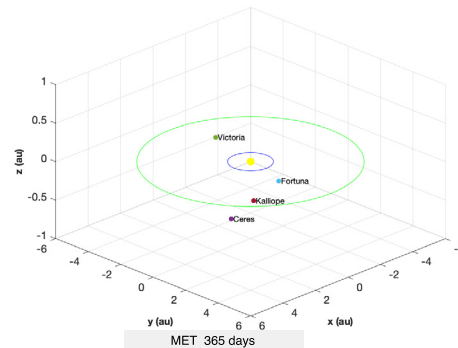
E.3. Asteroid orbits

Asteroid orbits can also be found from the JPL Horizons System. Another option is The Asteroid Orbital Elements Database [3]. An example from this database is shown in Example E.2. Orbit propagation in the plot uses Keplerian propagation.

```

1 asteroids = {'ceres' 'victoria' 'Fortuna' '
2             Kalliope'};
3 ReadAsteroidDatabase(asteroids, 'astorbshort.
4             dat');

```



Example E.2: Four asteroids showing their heliocentric orbits. The orbit propagation is Keplerian.

E.4. Planetary orientation

Planetary orientation is determined by three angles:

1. Rotation angle about the planet's axis of rotation;
2. Nutation of the axis of rotation;
3. Precession of the axis of rotation.

The Earth-fixed and Earth-centered inertial (ECI) coordinate frames are illustrated in Fig. E.1.

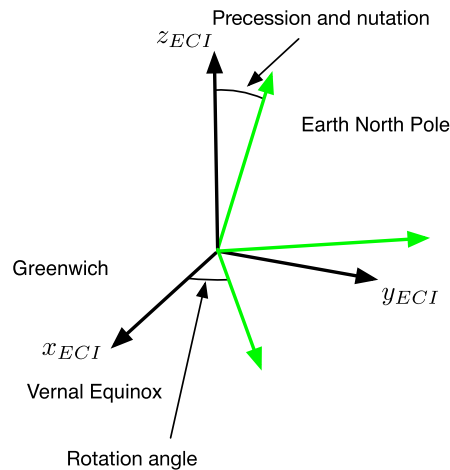


Figure E.1 ECI and Earth-fixed coordinate frames. In many applications, only the rotation angle needs be considered. The green arrows are the Earth-fixed frame.

The rotation angle is the daily rotation of the planet. This is the dawn-to-dusk cycle we experience on the Earth. Precession is the slow, gravity-induced rotation of the axis. The Earth's precession is a 26 000-year cycle. Nutation is the short-period oscillation of the plane's axis. The three angles determine the orientation of the planet with respect to the inertial frame. If a planet-pointing spacecraft is using an inertial reference, such as a star camera, it needs the transformation matrix generated by these three angles to know how to relate a location on the planet with the orientation measured with the star camera.

When the Earth's magnetic field is used for attitude determination, it is also necessary to transform the magnetic field into the inertial frame. Example E.3 shows the International Space Station orbit in the ECI and the Earth-fixed frames in three dimensions. The rotation of the Earth is evident from the orbit track.

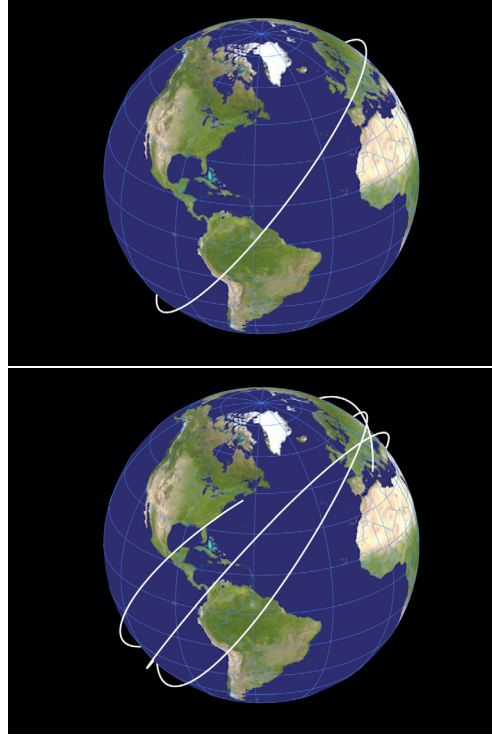
E.5. Asteroid dynamics

An asteroid's orientation may not fit into the rotation, precession, and nutation angle model of the major planets and moons. An asteroid may be rotating about all three axes in an irregular fashion. The angular rates may be coupled due to the inertia matrix of the asteroid not being that of a spherical body. For example, Example E.4 shows the rotation of the asteroid 4179 TOUTATIS [4]. The inertia matrix is normalized by I_{zz} . This does not change the dynamics since both terms are functions of I . The inertia of

```

1 %% Compute the ISSOrbit
2
3 [el, jD0] = ISSOrbit;
4 p         = Period(el(1));
5 t         = linspace(0,3*p,1000);
6 r         = RVOrbGen(el,t);
7
8 PlotOrbit(r,t,jD0)
9 PlotOrbit(r)

```



Example E.3: ISS orbit about the Earth in the ECI and Earth-fixed frames.

TOUTATIS cannot be accurately calculated using the shape and mass. It is believed that the mass distribution is irregular.

$$I\dot{\omega} - \omega^\times I\omega = 0 \quad (\text{E.1})$$

$$\frac{I\dot{\omega} - \omega^\times I\omega}{I_{zz}} = 0 \quad (\text{E.2})$$

The asteroid is subject to torques due to the Earth, Sun, and Moon's gravity. The disturbance torques are not included in the simulation. To study the effect of solar torques, the actual inertias, not ratios, are needed.

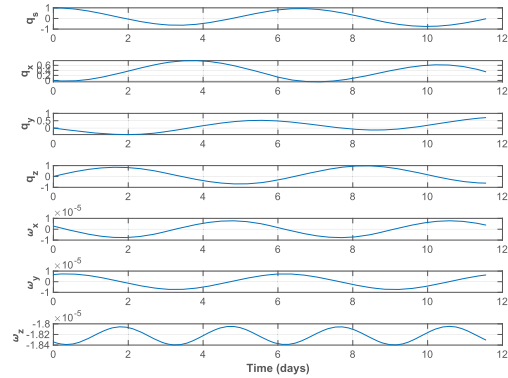
E.6. Stars

Star catalogs are needed for attitude determination using star cameras. There are many star catalogs. One well-known catalog is the Hipparcos catalog [5]. The European Space Agency's Hipparcos space astrometry mission catalogs over 100 000 stars. Example E.5 displays the catalog.

```

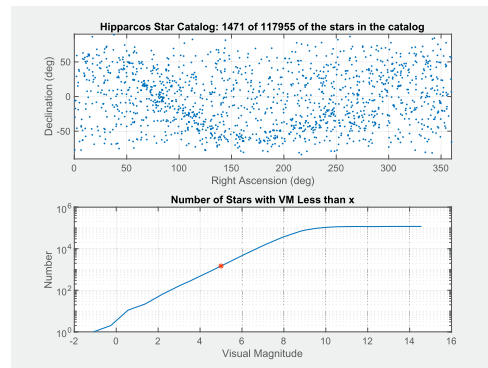
1
2 omega = [14.14;33.532;-90.793]*pi/180/86400;
3 x      = [1;0;0;0;omega];
4 n      = 1000;
5 xP     = zeros(7,n);
6
7 dT      = 1000;
8 for k = 1:n
9     xP(:,k) = x;
10    x = RK4(@RHS,x,dT,0);
11 end
12
13 yL = {'q_s' 'q_x' 'q_y' 'q_z' '\omega_x' '\omega_y' '\omega_z'};
14 TimeHistory((0:(n-1))*dT,xP,yL,'TOUTATIS');
15
16 function xDot = RHS(x,~)
17
18 inr = [ 3.0836 -7.1e-4 1.17e-3;...
19        -7.1e-4 3.235 1.3e-3;...
20        1.2e-3 1.3e-3 1];
21
22 omega = x(5:7);
23 q      = x(1:4);
24
25 qDot = QIToBDot(q,omega);
26 omegaDot = -inr\cross(omega,inr*omega);
27
28 xDot = [qDot;omegaDot];
29
30
31 end

```



Example E.4: 4179 TOUTATIS attitude motion. The motion is irregular. The inertia matrix is scaled by I_{zz} .

```
1 LoadCatalog
```



Example E.5: The Hipparcos star catalog. The bottom plot shows the number of stars as a function of visual magnitude. Visual magnitude is the brightness perceived by the human eye. Beyond ten the number of stars increases slowly.

When using the catalog for high-accuracy attitude determination, corrections must be made for:

1. radial velocity of the star;
2. parallax;
3. gravitational light deflection;

4. relativistic aberration due to planet and spacecraft velocity.

This process is known as stellar reduction [6].

A newer source for star data is the Gaia archive [7]. Gaia is a European space mission providing astrometry, photometry, and spectroscopy of more than a billion stars. It also has data for extragalactic and solar-system objects.

References

- [1] C. Acton, N. Bachman, B. Semenov, E. Wright, A look towards the future in the handling of space science mission geometry, in: *Enabling Open and Interoperable Access to Planetary Science and Helio physics Databases and Tools*, Planetary and Space Science 150 (2018) 9–12.
- [2] R.S. Park, HORIZONS System, Sep 2021.
- [3] T. Bowell, The asteroid orbital elements database, <http://naic.edu/~nolan/astorb.html>, Feb 2007.
- [4] Y. Takahashi, M.W. Busch, D.J. Scheeres, Spin state and moment of inertia characterization of 4179 Toutatis, *The Astronomical Journal* 146 (4) (sep 2013) 95.
- [5] M.A.C. Perryman, L. Lindegren, J. Kovalevsky, E. Hog, U. Bastian, P.L. Bernacca, M. Creze, F. Donati, M. Grenon, M. Grewing, F. van Leeuwen, H. van der Marel, F. Mignard, C.A. Murray, R.S. Le Poole, H. Schrijver, C. Turon, F. Arenou, M. Froeschle, C.S. Petersen, *The Hipparcos Catalogue*, *Astronomy & Astrophysics* 500 (1997) 501–504.
- [6] P.K. Seidelmann, *Explanatory Supplement to the Astronomical Almanac*, University Science Books, Mill Valley, CA, 1992.
- [7] Welcome to the Gaia Archive at ESA, <https://gea.esac.esa.int/archive/>.